Understanding Image Noise: Exploring Different Kinds of Noises in Image Processing

6 min read · Feb 4, 2025

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In today's digital world, images are everywhere — whether it's a selfie on your smartphone, a satellite capturing Earth, or an X-ray scan at a hospital. But before computers can process and analyze images effectively, they must overcome an unavoidable challenge: **noise**.

Noise distorts images, making it difficult for systems to detect objects, recognize patterns, and perform accurate analysis. This is a significant problem in **computer vision**, a field of artificial intelligence (AI) that enables machines to interpret and understand visual data. At the core of computer vision lies **image processing**, where images are enhanced, transformed, and analyzed to extract meaningful insights.

In this article, we'll explore what noise is, its sources, different types, and how to add noise using MATLAB.

What is Noise in Digital Images?

In simple terms, noise is any **unwanted variation** in an image that reduces its quality. It introduces random distortions, such as unwanted pixels or fluctuations in color and brightness. Mathematically, noise can be represented as:

$$I'(x, y) = I(x, y) + n(x, y)$$

Mathematical Representation of Noise

I′(x,y) is the noisy image,

I(x,y) is the original image,

n(x,y) is the noise function that adds distortion.

Sources of Noise in Digital Images

Noise in images is introduced at different stages, from capturing an image to processing and storing it. The most common sources of noise include:

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1. Capture Noise

This occurs when the image is being taken and can result from:

Lighting conditions; too much or too little light can introduce distortions.

Sensor issues; temperature fluctuations or electrical interference can cause random pixel variations.

Lens problems; dust, vibration, and focus errors affecting image clarity.

2. Sampling Noise

Since digital images represent real-world scenes in discrete pixel values, some details may be lost during conversion. This leads to aliasing effects, where fine details appear distorted.

3. Processing Noise

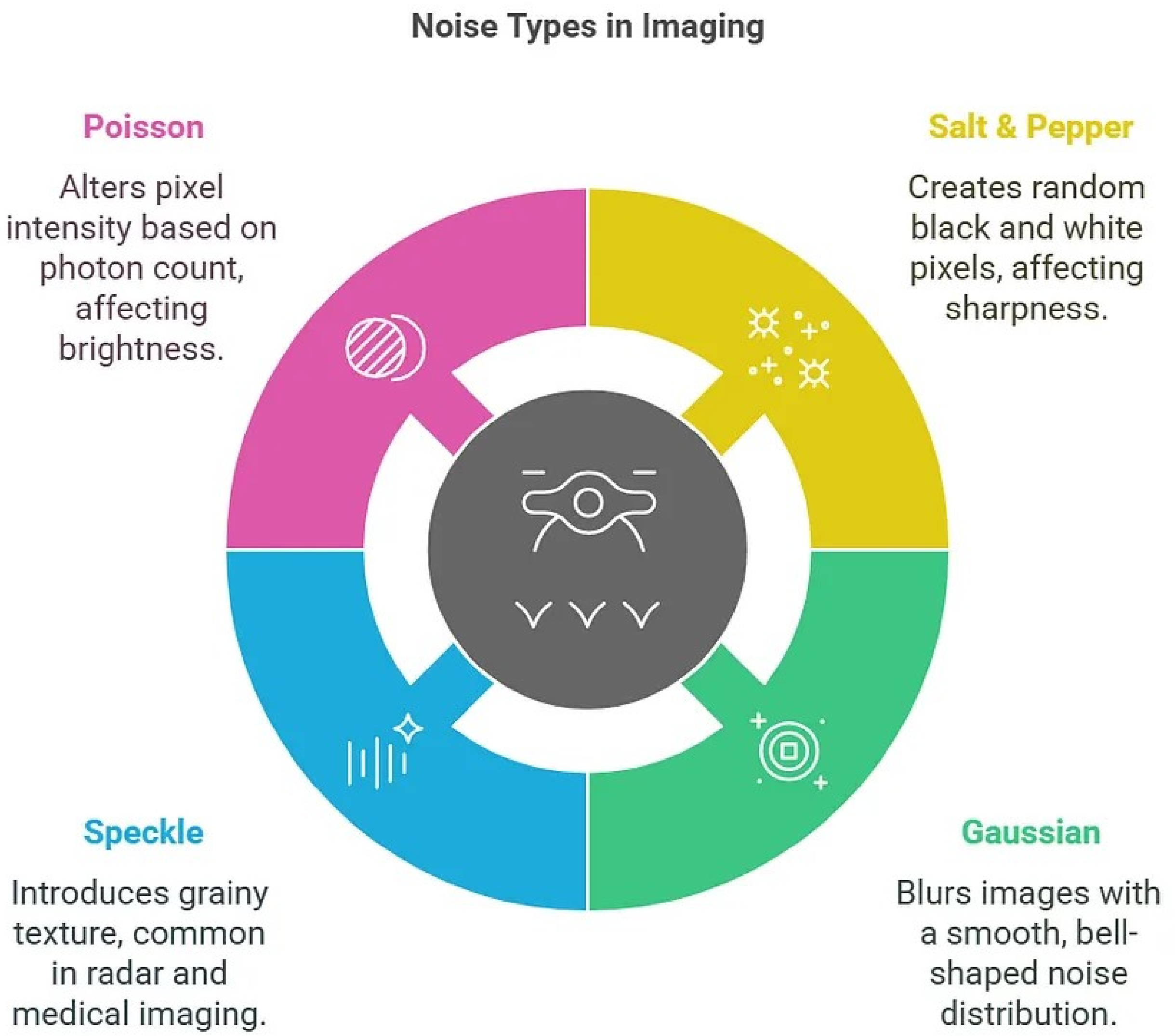
Errors can occur due to limitations in numerical precision (e.g., rounding errors, integer overflow) or mathematical approximations in image processing algorithms.

4. Image Compression Noise

Many image formats, like JPEG, use lossy compression to reduce file size by removing fine details. This process introduces compression artifacts that degrade image quality. Lossless formats like PNG avoid this problem but result in larger file sizes.

Characterization of Noise in Images

Noise affects images in different ways, and each type has unique characteristics. Following are some common types of noises.



Noise Types in Imaging

Salt-and-Pepper Noise

Salt-and-pepper noise also known as **Impulse Noise** appears as **random white and black pixels** scattered across the image, resembling grains of salt and pepper. It occurs due to:

Faulty camera sensors that introduce dead or stuck pixels.

Transmission errors in digital communication.

Bit errors during image storage and retrieval.

This type of noise is also known as **impulse noise** because it affects only specific pixels, leaving the rest of the image unchanged. It is easier to remove than other types of noise because it is localized and does not affect all pixels.

Salt-and-Pepper Noise using MATLAB

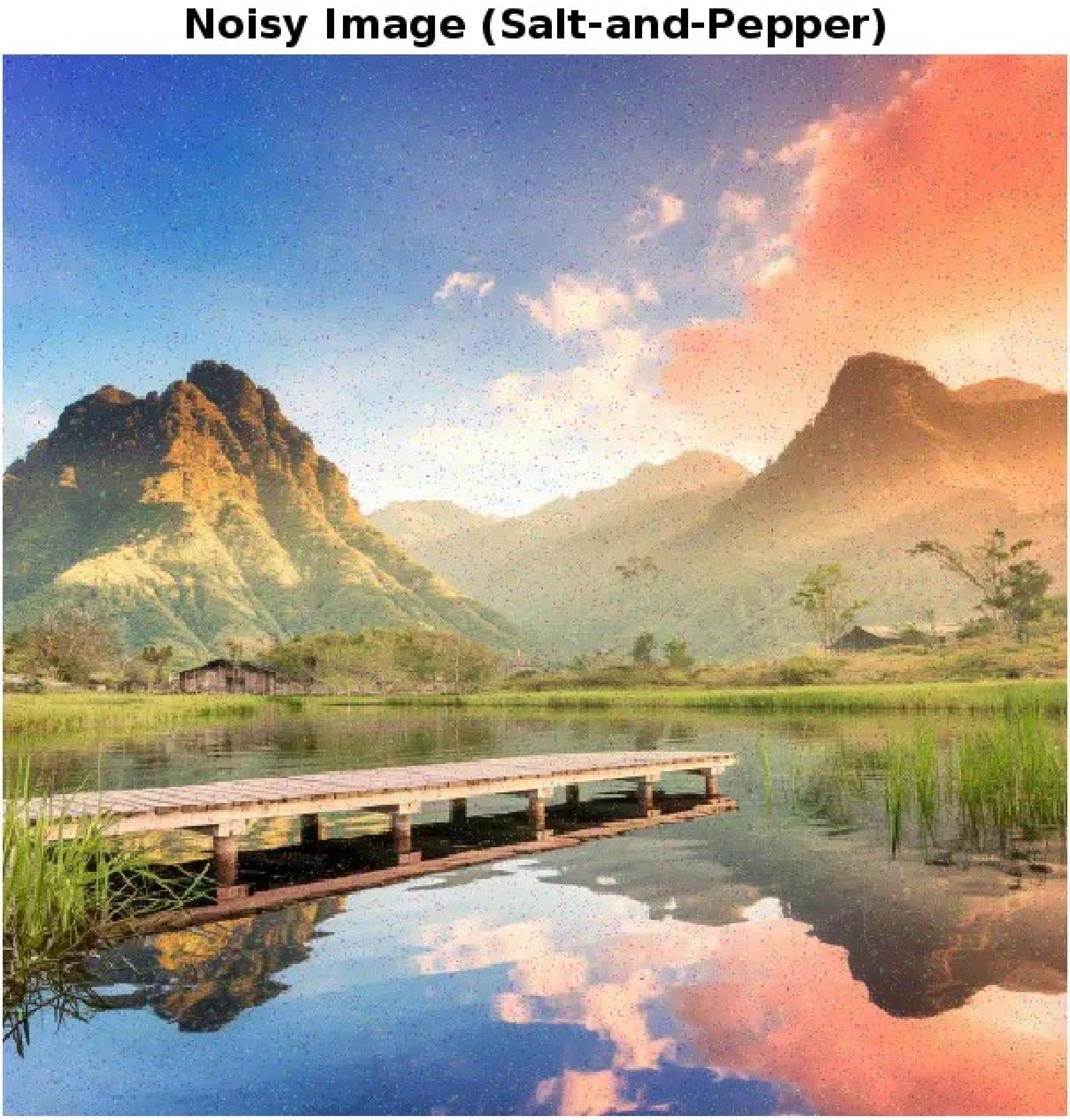
To begin, read and display the original image.

```
original_image = imread('Scenery Image.jpeg');
imshow(original_image);
title('Original Image');
```

MATLAB output; Original Image

Now, Adding Salt & Pepper Noise.

```
noisy_image = imnoise(original_image, 'salt & pepper', 0.02);
figure;
imshow(noisy_image);
title('Noisy Image (Salt-and-Pepper)');
```



MATLAB output; Salt & Pepper Noise

Gaussian Noise

Gaussian noise is the most common type of noise in digital images, caused by natural variations in lighting, sensor limitations, and electronic interference. Unlike salt-and-pepper noise, Gaussian noise is spread smoothly across the entire image rather than affecting only specific pixels.

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Gaussian noise follows a **normal distribution**, a common pattern in statistics where most values are centered around an average, with fewer values appearing as you move away from the center. This distribution forms a smooth, bell-shaped curve that is symmetrical, with its spread determined by how much values vary from the average. In the context of images, this means pixel values deviate randomly from their expected intensity, but most remain close to the mean.

The mathematical formula for Gaussian noise is:

$$\varphi(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(z-\mu)^2}{2\sigma^2}}$$

Mathematical Representation of Gaussian Noise

where:

μ is the mean (average noise level).

σ^2 is the variance (measures how spread out the noise is).

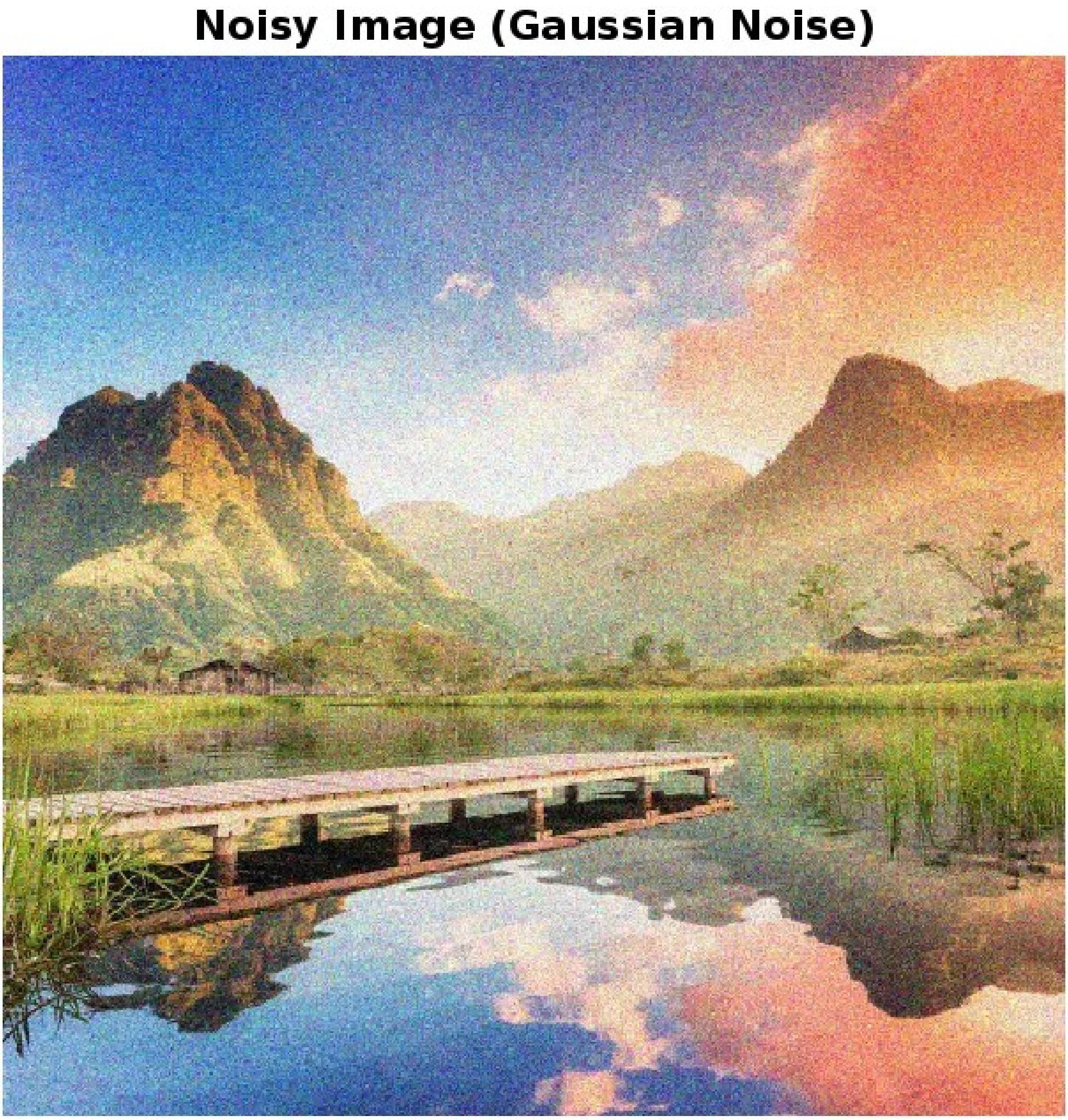
σ (standard deviation) is the square root of variance and determines the intensity of noise fluctuations.

> ***Higher variance*** *means stronger noise with more drastic pixel changes.* ***Lower variance*** *results in subtle noise, making the image appear smoother.*

Gaussian Noise using MATLAB

```
noisy_image_gaussian = imnoise(original_image, 'gaussian', 0, 0.03);
figure;
```

```
imshow(noisy_image_gaussian);
title('Noisy Image (Gaussian Noise)');
```



MATLAB output; Gaussian Noise

Speckle Noise

Speckle noise is a granular, high-frequency distortion that commonly affects radar, ultrasound, and medical imaging (MRI, CT scans). It is caused by interference between coherent waves reflected from multiple scatterers, resulting in a grainy texture that reduces image clarity.

Unlike other types of noise, speckle noise follows a multiplicative model, meaning its intensity is directly proportional to the pixel values in the image. This makes it particularly challenging to remove, as traditional noise reduction techniques may blur important details.

The speckle noise is modelled as multiplicative noise, defined as

$$I_o = I_f + N_m I_f + N_a$$

Mathematical Representation of Speckle Noise

If is a original image

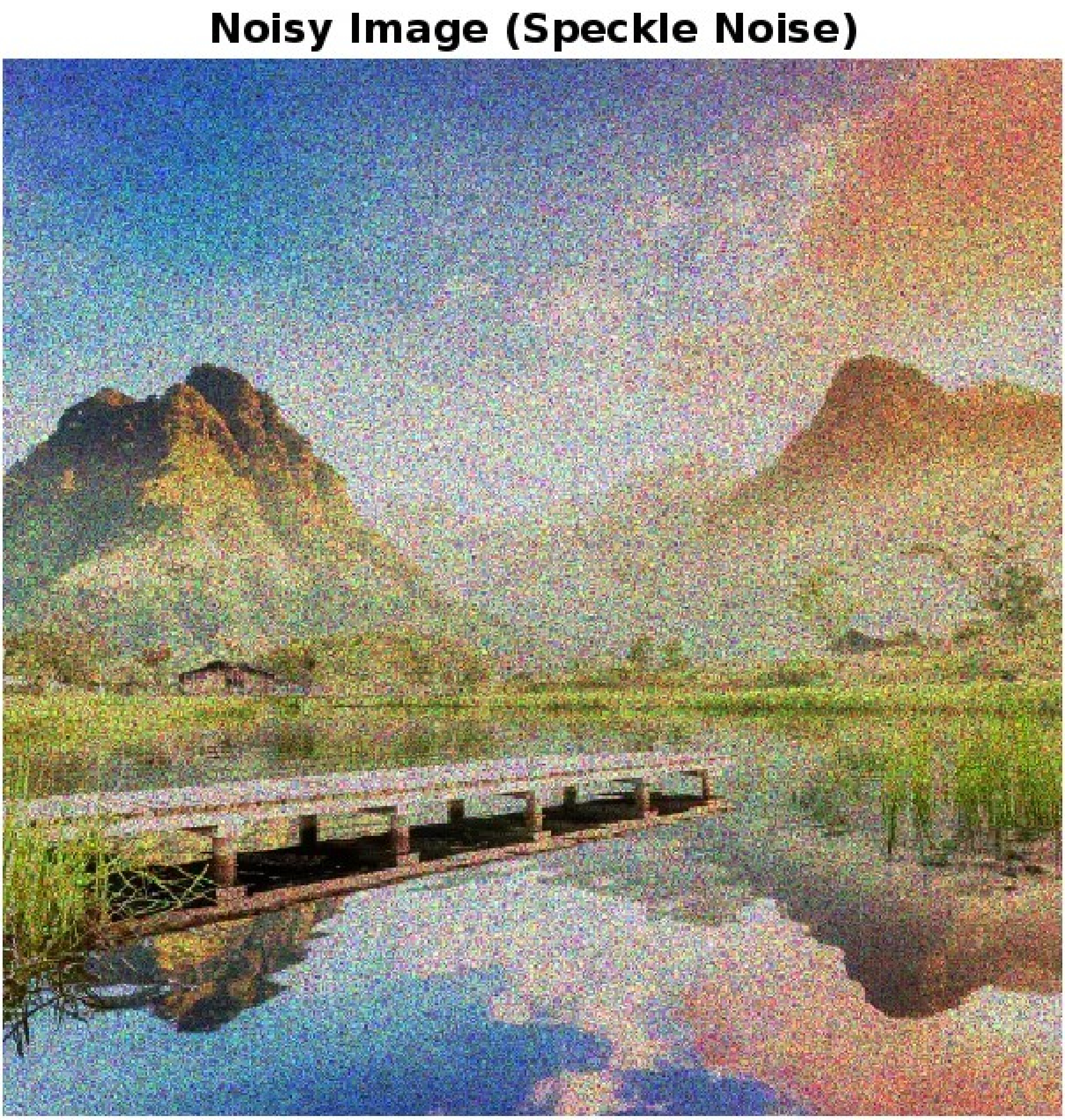
Io means observed image

Nm means multiplicative noise

Na is an additive noise

Speckle Noise using MATLAB

```
speckleNoise = imnoise(original_image, 'speckle', 0.3);
figure;
imshow(speckleNoise)
title('Noisy Image (Speckle Noise)');
```



MATLAB output; Speckle Noise

Poisson Noise

Poisson noise, also known as **shot noise**, happens due to random variations in the number of photons that reach a camera sensor during image capture. This noise depends on the brightness of the image, meaning that brighter areas have more noise, while darker regions have less.

Unlike Gaussian noise, Poisson noise is not simply added to an image — it naturally occurs because light is made up of individual photons arriving at different rates. This randomness follows a Poisson distribution, where the noise level increases as the light intensity increases. It is commonly seen in low-light images, medical scans, and astronomy pictures where fewer photons are available to create a clear image.

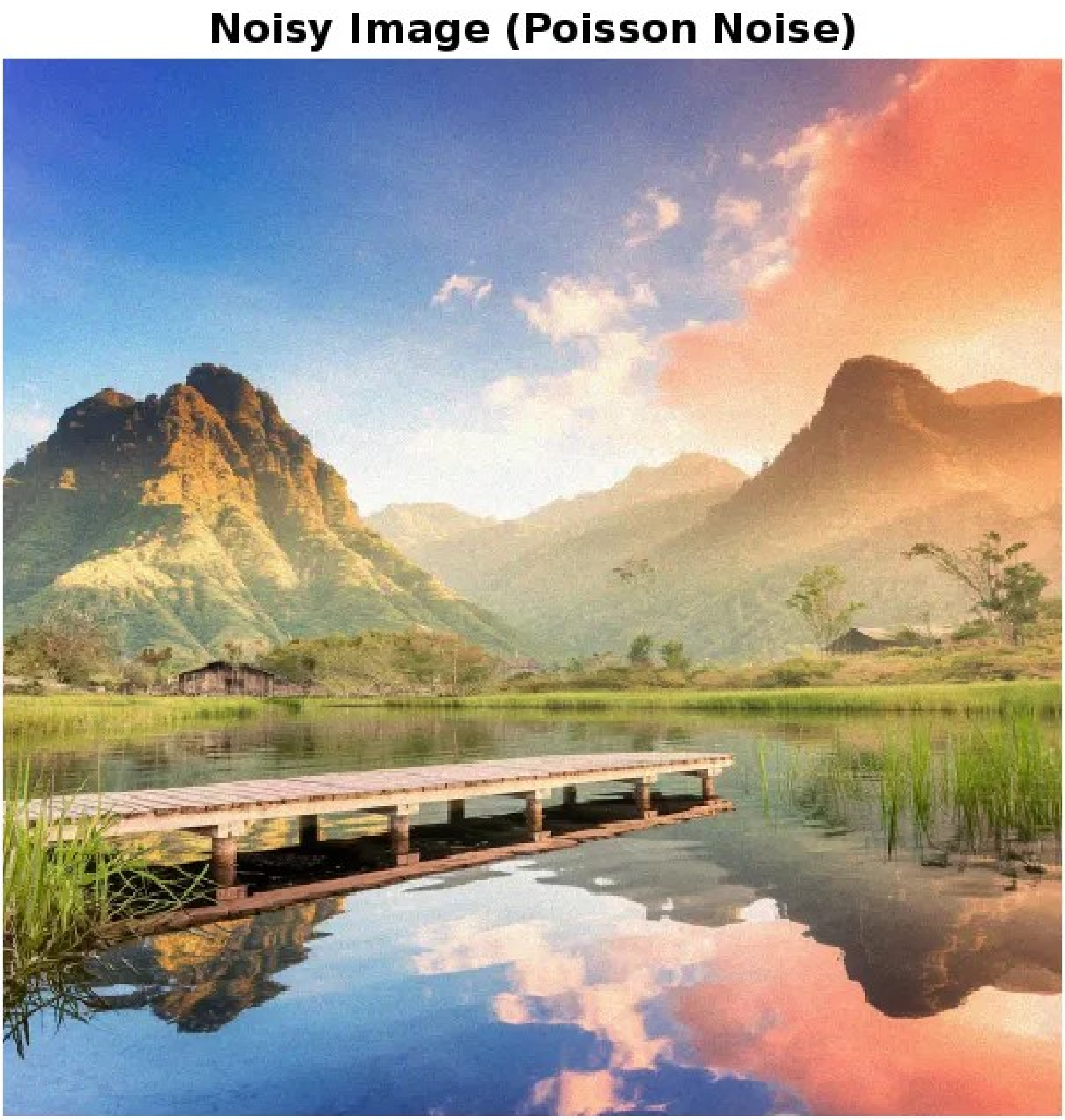
$$N(x, y) = P(I(x, y))$$

Mathematical Representation of Poisson Distribution

P(I(x,y)) is the Poisson-distributed noise added to the image.

Poisson Noise using MATLAB

```
poissonNoise = imnoise(original_image, 'poisson');
figure;
imshow(poissonNoise), title('Noisy Image (Poisson Noise)');
```



MATLAB output; Poisson Noise

Conclusion

Image noise is an inevitable challenge in image processing, affecting everything from medical scans to satellite images. Understanding different types of noise: **Gaussian, Salt & Pepper, Speckle, and Poisson** is crucial for developing effective noise reduction techniques. Each type of noise originates from different sources and impacts images in unique ways. While Gaussian noise results from electronic interference, Salt & Pepper noise appears due to transmission errors, Speckle noise arises in coherent imaging systems, and Poisson noise is caused by photon variations in low-light conditions.

By identifying the type of noise present, image processing techniques can be applied to enhance image quality, improve object detection, and ensure better analysis in fields like healthcare, surveillance, and remote sensing. Filtering methods like low-pass and Gaussian filters help reduce noise, making images clearer and more useful for further processing. Understanding and managing noise is a key step in advancing computer vision and AI-driven image analysis.

Computer Vision · Image Processing · Matlab

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Hello reader! I am a Computer Engineering student with a passion for Data Science & AI who loves to write about tech.

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